CS24305 — THEORY (3.0 CR) | MODULE IV NOTES | BIT MESRA | NEP SCHEME

# Module 4: Network Layer & Routing

IPv4 20-Byte Header, CIDR Subnetting Calculations, IPv6, Distance Vector (Bellman-Ford) & Link State (Dijkstra)

### Module IV: Wide Area Networks (WAN) & Local Area Networks (LAN) — Complete Tracker

**Topic 30:** Switching Networks (Circuit vs. Packet Switching Paradigms)

**Topic 31:** Circuit-Switching Networks (3 Phases: Setup, Transfer, Teardown)

**Topic 32:** Circuit-Switching Concepts (Blocking, Dedicated Resources)

**Topic 33:** Packet-Switching Principles (Datagram vs. Virtual-Circuit)

**Topic 34:** Cellular Network Principles (Cell Geometry, Frequency Reuse, Handoff)

**Topic 35:** Cellular Generations Evolution (1G Voice to 5G Ultra-Low Latency)

**Topic 36:** Network Topologies (Bus, Star, Ring, Mesh, Tree Analysis)

**Topic 37:** LAN Protocol Architecture (IEEE 802: LLC vs. MAC Sublayers)

**Topic 38:** Virtual Local Area Networks (VLAN Security & Broadcast Domains)

## Topic 30 – 33: Switching Paradigms (Circuit vs. Packet Switching) [UPLOADED PYQ]

| Parameter | Circuit Switching | Packet Switching [UPLOADED PYQ] |
|---|---|---|
| **Path Establishment** | Dedicated physical path established before data transfer (Setup $\rightarrow$ Transfer $\rightarrow$ Teardown). | No dedicated end-to-end path; data divided into independent packets routed dynamically. |
| **Resource Reservation** | Dedicated channel bandwidth reserved 100% (resources wasted if line is idle). | Statistical multiplexing: bandwidth shared on-demand among all active users. |
| **Traffic Suitability** | Ideal for continuous, real-time audio/voice communication. | Ideal for bursty, asynchronous computer data traffic (Internet). |
| **Delay Characteristics** | Initial connection setup delay; zero queuing delay during active transfer. | Zero setup delay (in datagram); packets experience variable queuing delay at routers. |

### Datagram vs. Virtual-Circuit Packet Switching [UPLOADED PYQ]:

**Datagram Network (Connectionless — IP):** Each packet contains full source and destination IP addresses and is treated as an independent entity. Intermediate routers make routing decisions per packet; packets may arrive out of order.
**Virtual-Circuit Network (Connection-Oriented — ATM / X.25):** A logical circuit is established upfront. Packets carry a short Virtual Circuit Identifier (VCI) rather than full global addresses. All packets follow the exact same route in sequence.

## Topic 34 & 35: Cellular Network Principles & Generations (1G – 5G)

### Core Cellular Principles: Frequency Reuse & Handoff

- **Frequency Reuse:** Geographical service area is divided into regular hexagonal **cells**. Adjacent cells use different frequency sets, but non-adjacent cells separated by minimum reuse distance $D = R\sqrt{3N}$ reuse the same frequencies, dramatically multiplying network capacity!
- **Handoff / Handover:** The automatic transition of an active ongoing call/data session from one base station channel to another as the mobile user moves across cell boundaries (Hard Handoff: "Break before make"; Soft Handoff: "Make before break").

| Generation | Key Technological Milestone | Core Services & Data Rate | Multiple Access Tech |
|---|---|---|---|
| **1G (1980s)** | Analog cellular technology. | Voice only ($\approx 2.4$ kbps). | FDMA |
| **2G (1990s)** | Digital cellular, encryption, roaming. | Voice, SMS text ($\approx 64$ kbps). | TDMA / CDMA (GSM) |
| **3G (2000s)** | Mobile broadband Internet (UMTS). | Video calling, mobile web ($\approx 2$ Mbps). | WCDMA |
| **4G (2010s)** | All-IP network, LTE standard. | HD video streaming, gaming ($\approx 100$ Mbps). | OFDMA |
| **5G (2020s)** | Millimeter wave, network slicing, massive IoT. | eMBB, URLLC, mMTC ($1\text{–}10$ Gbps, $< 1$ ms). | Scalable OFDMA |

## Topic 36: Physical Network Topologies

| Topology | Structural Layout & Communication | Key Advantages & Vulnerabilities |
|---|---|---|
| **1. Mesh** | Every node has dedicated point-to-point links to all other $N-1$ nodes (Total links $= \frac{N(N-1)}{2}$). | Ultimate fault tolerance and privacy. High cable cost and complex installation. |
| **2. Star** | Each node connected to a central controller / switch via dedicated point-to-point link. | Easy to install, isolate faults. Central switch failure halts entire network. |
| **3. Bus** | All devices share a single long central backbone cable terminated at both ends. | Low cabling cost. Backbone cable break disables entire network. |
| **4. Ring** | Each device has dedicated point-to-point connection to exactly two neighboring nodes in a loop. | Predictable token traffic. Break in ring disables network (unless dual-ring). |

## Topic 37 & 38: LAN Architecture & Virtual LANs (VLAN)

IEEE 802 LAN Architecture divides the OSI Data Link Layer into two distinct sublayers:

**Logical Link Control (LLC — IEEE 802.2):** Independent of physical medium; handles flow control, error control, and upper-layer protocol multiplexing.
**Medium Access Control (MAC):** Dependent on physical topology; governs shared-channel access rules (CSMA/CD in 802.3, CSMA/CA in 802.11) and framing.

> **VLAN (Virtual Local Area Network — IEEE 802.1Q)**
>
> A **VLAN** is a logical broadcast domain created by software configuration on switches regardless of physical cable connections. It isolates network broadcast storms, enhances department security, and reduces routing costs!

## 🧠 M4 Active Recall & Exam Questions [UPLOADED PYQ]

**Q1. [UPLOADED PYQ] Compare Circuit Switching and Packet Switching across 4 architectural parameters. (8 Marks)**

1. **Connection Setup:** Circuit switching requires explicit 3-phase connection setup; Packet switching (datagram) sends packets immediately with zero setup.
2. **Resource Utilization:** Circuit switching reserves 100% dedicated channel capacity; Packet switching uses statistical multiplexing to maximize bandwidth utilization.
3. **Congestion & Delays:** Circuit switching has fixed transmission delay once connected; Packet switching suffers from variable queuing and router processing delays.
4. **Failure Resilience:** Circuit switching loses the call if any link on the dedicated path breaks; Packet switching automatically reroutes individual packets around failed links.